

Group Topic: *Predicting Student Dropout Risk Using Academic and Behavioral Data*

Lab No.

Git Repo/ Colab Link:

Date:

<https://github.com/dgbb-prog/student-dropout-bigdata-datacleaning>

Objectives:

The goal of this exercise is to develop and train various machine learning models that will predict the likelihood of students dropping out based on their academic and behavioral performance. This exercise centers on creating and comparing basic and sophisticated classifiers to evaluate their predictive accuracy and produce metrics results.

Specifically, the activity aims to:

- Implement Logistic Regression as the baseline model
- Train Random Forest and XGBoost classifiers
- Apply tuned hyperparameters for XGBoost optimization
- Compare algorithm performance using evaluation metrics
- Generate a metrics summary table containing Accuracy, F1-Score, and ROC-AUC scores

Tasks Completed:

The following tasks were completed as part of the ***Model Training (Algorithms) role:***

1. Defined and Implemented Machine Learning Models

Implemented three classification algorithms for predicting student dropout risk: Logistic Regression, Random Forest, and Tuned XGBoost.

Code Used:

```
models = {
    'Logistic Regression': LogisticRegression(random_state=42, max_iter=1000),
    'Random Forest'      : RandomForestClassifier(random_state=42, n_estimators=200),
    'XGBoost (Tuned)'    : XGBClassifier(
        random_state=42, eval_metric='logloss',
        learning_rate=0.1, max_depth=5,
        n_estimators=200, subsample=1.0
    )
}
```

2. Trained the Machine Learning Models

Trained each classification model using the training dataset and generated predictions for the testing dataset.

Code Used:

```
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    y_proba = model.predict_proba(X_test)[:, 1] if hasattr(model, 'predict_proba') else None
```

3. Evaluated Model Accuracy

Calculated and recorded the accuracy score of each trained algorithm to compare prediction performance.

Code Used:

```
acc = accuracy_score(y_test, y_pred)
results[name] = {
    'model' : model,
    'predictions': y_pred,
    'proba' : y_proba,
    'accuracy' : acc,
}
print(f"{'✅'} {name:<28} Accuracy: {acc:.4f}")
```

4. Generated Algorithm Metrics Summary Table

Compiled a performance summary table containing Accuracy, Precision, Recall, F1-Score, and ROC-AUC scores for all trained models.

Code Used:

```
from sklearn.metrics import roc_auc_score

summary_rows = []
for name, data in results.items():
    report = classification_report(y_test, data['predictions'], output_dict=True)
    auc_score = roc_auc_score(y_test, data['proba']) if data['proba'] is not None else None
    summary_rows.append({
        'Model' : name,
        'Accuracy' : f"{data['accuracy']:.4f}",
        'Precision (1)' : f"{report['1']['precision']:.4f}",
        'Recall (1)' : f"{report['1']['recall']:.4f}",
        'F1-Score (1)' : f"{report['1']['f1-score']:.4f}",
        'ROC-AUC' : f"{auc_score:.4f}" if auc_score else '-',
    })

summary_df = pd.DataFrame(summary_rows).set_index('Model')
print("\n📊 Algorithm Performance Summary")
print("=" * 70)
print(summary_df.to_string())
```

Results and Analysis

All machine learning classifiers were successfully trained and evaluated using student academic and behavioral data set. There were three classification algorithms implemented: Logistic Regression, Random Forest and Tuned XGBoost.

In this project, Logistic Regression was used as a base model since it was able to give more stable predictions about student behavior in comparison with other models. Prediction performance of this classifier was increased by using Random Forest algorithm which was based on ensemble learning and consisted of 200 estimators.

Of all classifiers implemented during the project, Tuned XGBoost gave the best result because of optimal parameters settings which included learning rate, max depth and number of estimators. Also this model showed better prediction performance in comparison with other algorithms.

Metrics table was made which contains Accuracy, Precision, Recall, F1-Score and ROC-AUC values of classifiers' prediction performances. This table helped to evaluate effectiveness of machine learning models during student dropout risk prediction process.

During this activity it was shown that application of advanced ensemble methods like Random Forest and XGBoost can be more efficient than classical machine learning algorithms.

Challenges and Solutions

1. Challenge:

Some models initially produced lower prediction accuracy during training.

Solution:

Implemented multiple classification algorithms and compared their performance using evaluation metrics.

2. Challenge:

The XGBoost model required proper parameter tuning for better performance.

Solution:

Applied tuned hyperparameters such as learning rate, max depth, and number of estimators to improve model accuracy.

3. Challenge:

Difficulty comparing the performance of multiple algorithms.

Solution:

Generated a metrics summary table containing Accuracy, Precision, Recall, F1-Score, and ROC-AUC results for easier comparison.

Conclusion

All the objectives set for the activity were met successfully through training multiple machine learning models to predict student dropouts.

The activity enhanced knowledge on the training of models, comparisons of different algorithms, and their performance evaluation by making use of Python, Scikit-learn, and XGBoost packages. Logistic Regression was a good base model, but the Random Forest and Tuned XGBoost performed much better.

Metrics summary generated from the activity helped determine the best model and the need for selecting and tuning the right model for predictive analysis.